MATERIAL MODELING 9

CS 184: FOUNDATIONS OF COMPUTER GRAPHICS

1 BRDF Physical Plausibility

A physically plausible BRDF must satisfy certain properties. Consider the following BRDF:

$$f_r(\omega_i, \omega_o) = \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_i \cdot \mathbf{n})(\omega_o \cdot \mathbf{n})}$$

where k and n are constants, ω_i is the incoming direction, ω_o is the outgoing direction, and $\mathbf{n}$ is the surface normal.

1. Check if this BRDF satisfies positivity.

2. Check if this BRDF satisfies Helmholtz reciprocity (symmetry).

3. Find the constraints on k and n that would make this BRDF conserve energy.

2 Cook-Torrance BRDF Calculation

This problem will walk you through a full shading calculation for a single point on a surface using the Cook-Torrance microfacet model.

Scenario:

- A point on a surface has normal $\mathbf{n} = (0, 0, 1)$.
- The viewer is in direction $\omega_o = \left(\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right)$.
- A single point light provides radiance L_i from direction $\omega_i = \left(-\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right)$.
- Assume for this setup, the material properties evaluate to $D = 1/\pi$, $G = 1$, and $F = F_0$.

Your task is to calculate the final outgoing radiance, $L_o(\omega_o)$.

1. Calculate the halfway vector, h.

2. Calculate the dot products $(\mathbf{n} \cdot \omega_i)$ and $(\mathbf{n} \cdot \omega_o)$.

3. Assemble the full Cook-Torrance BRDF value, $f_r = \frac{D \cdot G \cdot F}{4(\mathbf{n} \cdot \omega_i)(\mathbf{n} \cdot \omega_o)}$.

4. Calculate the final outgoing radiance L_o using the reflection equation for a single point light: $L_o = f_r \cdot L_i \cdot (\mathbf{n} \cdot \omega_i)$.